# Report on Inference for Empirical Wasserstein Distances on Finite Spaces by Sommerfeld and Munk (2018)

Steve Broll (sb2643)

May 2021

## 1 Summary and Review

### 1.1 Overview

Sommerfeld and Munk [1] derive limit properties for the Wasserstein distance for finitely supported probability measures, discuss applications and limitations, and run simulations and data analysis to verify the limit properties and demonstrate applications.

The first section is a review of Wasserstein distance for both infinitely and finitely supported probability measures, then provides an overview of the main results, which will be discussed more in the next section. The authors note that $n$ out of $n$ bootstrap is not consistent for Wasserstein distance, but $m$ out of $n$ bootstrap and a new scheme, denoted directional bootstrapping by the authors, are consistent for Wasserstein Distance.

The second section shows the main results, which are the distributional limits of Wasserstein distance for finitely supported probability measures, and their proofs. The proofs rely on showing the Hadamard directional derivatives, which are an extension of Hadamard derivatives, for Wasserstein distance. The downside of the 'directional' part is that naive $n$ out of $n$ bootstrapping requires Hadamard differentiability, not Hadamard directional differentiability, and so is not consistent as mentioned in the first section.

The third section compares simulated results with the distributional limits, and test hypotheses using Wasserstein distance and other methods for fingerprint and microbial datasets. The simulated distributions were decent approximations for the limiting distribution,even at $n = 50$, shown by distribution plots and by small KS-statistics. In the data analysis section, Wasserstein distance based tests performed as well as 'state of the art' methods for the fingerprint data, and confidence intervals were constructed for the microbial data with computational cost that is noted to be quite small.

The fourth and final section discusses other work connecting the main limit results to results from other papers and from other areas like transport distances. In particular, for a continuous distribution setting when $d = 1$ and $p = 2$ the scaling rate is known to be $n^{1/2}$, but we do not have a known scaling rate for $d \geq 2$. The supplementary material for the paper is very useful for constructing the full proofs and for a detailed description of the author's directional bootstrapping scheme.

### 1.2 Main Results

For the main theorem of the paper, we consider the scenario where $\mathcal{X} = \{x_1, \ldots, x_N\}$ is a **finite** space, and so every probability measure on $\mathcal{X}$ is a finitely supported vector. So the Wasserstein distance of order $p$ between finitely supported probability measures $\boldsymbol{r}, \boldsymbol{s}$ is

$$W_p(\boldsymbol{r}, \boldsymbol{s}) = \left\{ \min_{w \in \prod(\boldsymbol{r}, \boldsymbol{s})} \sum_{x, x' \in \mathcal{X}} d^p(x, x') w_{x,x'} \right\}^{1/p}$$

Theorem 1 of the paper states that for $\hat{\mathbf{r}}_\mathbf{n}$ and $\hat{\mathbf{s}}_\mathbf{m}$ generated by $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathbf{r}$ and $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathbf{s}$ we have convex sets

$$\Phi_p^* = \{\mathbf{u} \in \mathbb{R}^\chi : \ u_x - u_{x'} \leq d^P(x, x'), \quad x, x' \in \chi \ \},$$

$$\Phi_p^*(\mathbf{r},\mathbf{s}) = \left\{ (\mathbf{u},\mathbf{v}) \in \mathbb{R}^\chi \times \mathbb{R}^\chi : \begin{array}{c} \langle \mathbf{u},\mathbf{r}\rangle + \langle \mathbf{v},\mathbf{s}\rangle = \mathbf{W_p^p}(\mathbf{r},\mathbf{s}), \\ u_x + v_{x'} \le d^p(x,x'),\ x,x' \in \chi \end{array} \right\}$$

$$\text{and Multinomial Covariance } \Sigma(\mathbf{r}) = \begin{bmatrix} r_{x_1}(1-r_{x_1}) & -r_{x_1}r_{x_2} & \cdots & -r_{x_1}r_{x_N} \\ -r_{x_2}r_{x_1} & r_{x_2}(1-r_{x_2}) & \cdots & -r_{x_2}r_{x_N} \\ \vdots & \vdots & \ddots & \vdots \\ -r_{x_N}r_{x_1} & -r_{x_N}r_{x_2} & \cdots & r_{x_N}(1-r_{x_N}) \end{bmatrix} \text{ s.t.}$$

for independent Gaussian $\mathbf{G} \sim N(0,\Sigma(\mathbf{r}))$, $\mathbf{H} \sim N(0,\Sigma(\mathbf{s}))$ we have four main results:

1. One Sample - Null Hypothesis
$$n^{1/(2p)} W_p(\hat{\boldsymbol{r}}_{\boldsymbol{n}}, \boldsymbol{r}) \to (\max_{\boldsymbol{u}\in\Phi_p^*} \langle \boldsymbol{G},\boldsymbol{u}\rangle)^{1/p} \text{ as } n \to \infty$$

2. One Sample - Alternative Hypothesis
$$n^{1/2}\{W_p(\hat{\boldsymbol{r}},\boldsymbol{s}) - W_p(\hat{\boldsymbol{r}}_{\boldsymbol{n}},\boldsymbol{s})\} \to \frac{1}{p} W_p^{1-p}(\boldsymbol{r},\boldsymbol{s})(\max_{\boldsymbol{u},\boldsymbol{v}\in\Phi_p^*} \langle \boldsymbol{G},\boldsymbol{u}\rangle) \text{ as } n \to \infty$$

3. Two Samples - Null Hypothesis
$$\rho_{n,m}^{1/p} W_p(\hat{\boldsymbol{r}}_{\boldsymbol{n}}, \hat{\boldsymbol{s}}_{\boldsymbol{m}}) \to (\max_{\boldsymbol{u}\in\Phi_p^*} \langle \boldsymbol{G},\boldsymbol{u}\rangle)^{1/p}$$
$$\text{as } n,m \to \infty \text{ s.t. } n \wedge m \to \infty \text{ and } m/(n+m) \to \lambda \in [0,1]$$

4. Two Samples - Alternative Hypothesis
$$\rho_{n,m}^{1/p}\{W_p(\hat{\boldsymbol{r}}_{\boldsymbol{n}}, \hat{\boldsymbol{s}}_{\boldsymbol{m}}) - W_p(\boldsymbol{r},\boldsymbol{s})\} \to \frac{1}{p} W_p^{1-p}(\boldsymbol{r},\boldsymbol{s}))\left\{ \max_{\boldsymbol{u},\boldsymbol{v}\in\Phi_p^*} \sqrt{\lambda}\langle \boldsymbol{G},\boldsymbol{u}\rangle + \sqrt{1-\lambda}\langle \boldsymbol{H},\boldsymbol{v}\rangle \right\}$$
$$\text{as } n,m \to \infty \text{ s.t. } n \wedge m \to \infty \text{ and } m/(n+m) \to \lambda \in [0,1]$$

Note that these results are valid regardless of the dimension of the underlying space, as long as the probability measures have finite support. A full proof of the One Sample Null Hypothesis result is found at the end of this report.

### 1.3 Novelty

The novelty of the paper stems from the derived distributional limits for the finite case. Previous work on Wasserstein distance for continuous probability measures has shown that, as stated previously, the scaling factor for $d = 1, p = 2$ is $\sqrt{n}$ under the alternative and under the null under some tail regularity conditions. However, for $d \ge 2$ previous work seems to show that we can't find such a universal scaling factor.

The focus of most of the text and all of the graphs of the paper revolves around deriving and verifying these limits. The real data examples show that one application of these novel results, multivariate hypothesis testing based on Wasserstein distance, holds up to the performance of standard methods. There are some scenarios where the distance matrix may make the sensitivity or specificity drastically different from, for example, the chi-square test. If the distance matrix, underlying grid, and bins are chosen appropriately, tests based on the Wasserstein distance seem promising.

## 2 Proof of Theorem 1 part a

**Theorem 1.** *Let $p \ge 1$, $\mathbf{r},\mathbf{s} \in \mathcal{P}_\chi$ and $\hat{\mathbf{r}}_\mathbf{n}$ and $\hat{\mathbf{s}}_\mathbf{m}$ generated by IID samples $X_1,...,X_n \sim \mathbf{r}$ and $Y_1,...,Y_m \sim \mathbf{s}$ respectively. We define the convex sets*

$$\Phi_p^* = \{\mathbf{u} \in \mathbb{R}^\chi : \ u_x - u_{x'} \le d^P(x,x'), \quad x,x' \in \chi \ \},$$

$$\Phi_p^*(\mathbf{r},\mathbf{s}) = \left\{ (\mathbf{u},\mathbf{v}) \in \mathbb{R}^\chi \times \mathbb{R}^\chi : \begin{array}{c} \langle \mathbf{u},\mathbf{r}\rangle + \langle \mathbf{v},\mathbf{s}\rangle = \mathbf{W_p^p}(\mathbf{r},\mathbf{s}), \\ u_x + v_{x'} \leq d^p(x,x'),\ x,x' \in \chi \end{array} \right\}$$

*and the multinomial covariance matrix*

$$\Sigma(\mathbf{r}) = \begin{bmatrix} r_{x_1}(1-r_{x_1}) & -r_{x_1}r_{x_2} & \cdots & -r_{x_1}r_{x_N} \\ -r_{x_2}r_{x_1} & r_{x_2}(1-r_{x_2}) & \cdots & -r_{x_2}r_{x_N} \\ \vdots & \vdots & \ddots & \vdots \\ -r_{x_N}r_{x_1} & -r_{x_N}r_{x_2} & \cdots & r_{x_N}(1-r_{x_N}) \end{bmatrix}$$

*such that with independent Gaussian random variables* $\boldsymbol{G} \sim \mathcal{N}\{0,\Sigma(\boldsymbol{r})\}$, $\boldsymbol{H} \sim \mathcal{N}\{0,\Sigma(\mathbf{s})\}$ *we have the following result. (One sample-null hypothesis): with the sample size* $n$ *approaching* $\infty$*, we have the weak convergence*

$$n^{1/2}\{W_p(\hat{\mathbf{r}}_\mathbf{n},\mathbf{s}) - W_p(\mathbf{r},\mathbf{s})\} \xrightarrow{w} \frac{1}{p} W_P^{1-p}(\mathbf{r},\mathbf{s})(\max_{(\mathbf{u},\mathbf{v})\in\Phi_p^*(\mathbf{r},\mathbf{s})} \langle \boldsymbol{G},\mathbf{u}\rangle)$$

**Proof:**

$n\hat{\mathbf{r}}_\mathbf{n}$ is a sample of size $n$ from a multinomial distribution with probabilities $\mathbf{r}$.

From Wasserman's (2011) [2] proofs of Theorem 14.5 and Theorem 14.6, the log-likelihood of $p$ for a Multinomial$(n,p)$ is

$$l(p) = \sum_{j=1}^{k} X_j \log p_j$$

.

When we maximize $l$ we have to enforce the constraint that $\sum_j p_j = 1$. Using Lagrange multipliers, we instead maximize

$$A(p) = \sum_{j=1}^{k} X_j \log p_j + \lambda(\sum_j p_j - 1)$$

$$\frac{\partial A(p)}{\partial p_j} = \frac{X_j}{p_j} + \lambda$$

Setting the left side to 0, we hae $\hat{p}_j = -X_j/\lambda$, and with the constraint $\sum_j \hat{p}_j = 1$, $\lambda = n$. So $\hat{p}_j = X_j/n$.

For the variance of the MLE, we can directly compute the variance matrix of $\hat{p}$.

$$Var(\hat{p}) = Var(X/n) = n^{-2}Var(X) \implies Var(\hat{p}) = \frac{1}{n}\Sigma$$

$$\text{where } \Sigma = \begin{bmatrix} p_1(1-p_1) & -p_1p_2 & \cdots & -p_1p_k \\ -p_2p_1 & p_2(1-p_2) & \cdots & -p_2p_k \\ \vdots & \vdots & \ddots & \vdots \\ -p_kp_1 & -p_kp_2 & \cdots & p_k(1-p_k) \end{bmatrix}$$

For large $n$, $\hat{p}$ has approximately a multivariate Normal distribution. That is, as $n \to \infty$,

$$\sqrt{n}(\hat{p} - p) \xrightarrow{w} N(0,\Sigma)$$

So $(\hat{\mathbf{r}}_\mathbf{n} - \mathbf{r})\sqrt{n} \xrightarrow{w} \boldsymbol{G}$ as $n \to \infty$.

The Hadarmard derivative of the map $(\mathbf{r},\mathbf{s}) \mapsto W_p^p(\mathbf{r},\mathbf{s})$ from theorem 4

$$(\mathbf{h_1},\mathbf{h_2}) \mapsto \max_{(\mathbf{u},\mathbf{v})\in\Phi_p^*(\mathbf{r},\mathbf{s})} -(\langle \mathbf{u},\mathbf{h_1}\rangle + \langle \mathbf{v},\mathbf{h_2}\rangle)$$

can now be used with the delta method from Theorem 3 . Theorem 3 states that if $f$ is a function defined on a subset $F$ of $\mathbb{R}^d$ with values in $\mathbb{R}$ s.t.

- $f$ is Hadamard directionally differentiable at $\mathbf{u} \in F$ with derivative $f'_{\mathbf{u}} : F \to \mathbb{R}$ and
- there is a sequence of $\mathbb{R}^d$-valued random variables $X_n$ and a sequence of non-negative $\rho_n$ s.t. $\rho_n(X_n - \mathbf{u}) \xrightarrow{w}$ for some random variable $X$ taking values in $F$

Then $\rho_n\{f(X_n) - f(\mathbf{u})\} \xrightarrow{w} f_{\mathbf{u}}^{'(X)}$
The proof comes from Römisch (2004) [3] and goes as follows:
For each $n$ we define $F_n := \{h \in \mathbb{R}^d : \mathbf{u} + \frac{h}{\rho_n \in F}\}$ and $g_n : D_n \to \mathbb{R}, g_n(h) := \rho_n(f(\mathbf{u} + \frac{h}{\rho_n}) - f(\mathbf{u})$ for all $h \in F_n$.
Then we conclude from the Hadamard directional differentiability of $f$ that for every subsequence $h_{n'} \in D_{n'}$ converging to h, the sequence $g_{n'}$ satisfies $g_{n'}(h_{n'}) \to f'_{\mathbf{u}}(h)$.
In addition, $f'_{\mathbf{u}}(X)$ is Borel measurable, so the extended continuous mapping theorem for weak convergence implies that

$$g_n(\rho_n(X_n - \mathbf{u})) = \rho_n(f(X_n) - f(\mathbf{u}) \xrightarrow{w} f'_{\mathbf{u}}(X)$$

and we have our desired result.

Using the previously shown properties together with representation

$$\Phi_P^*(\mathbf{r}, \mathbf{r}) = \Phi_p^* \times (-\Phi_p^*).$$

We now have

$$W_p^p(\hat{\mathbf{r}}_\mathbf{n}, \mathbf{r})\sqrt{n} \xrightarrow{w} \max_{(\mathbf{u},\mathbf{v}) \in \Phi_p^*(\mathbf{r},\mathbf{r})} - \langle \mathbf{u}, \boldsymbol{G} \rangle \overset{D}{\sim} \max_{(\mathbf{u},\mathbf{v}) \in \Phi_p^*} \langle \mathbf{u}, \boldsymbol{G} \rangle$$

Applying the continuous mapping theorem with map $t \mapsto t^{1/p}$, we have our result. $\square$

# 3 References

# References

[1] Max Sommerfeld and Axel Munk. Inference for empirical wasserstein distances on finite spaces. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 80(1):219–238, 2018.

[2] Larry Wasserman. *All of statistics: a concise course in statistical inference.* Springer, 2011.

[3] Werner Römisch. Delta method, infinite dimensional. *Encyclopedia of Statistical Sciences*, 2004.